CUSTOMER CHURN ANALYSIS

Objective

The objective of this project is to analyse customer churn behaviour on a digital platform using key behavioural and demographic metrics. The analysis aims to identify which factors—such as login device, satisfaction score, tenure, coupon usage, order activity, app usage time, and gender—are most strongly associated with customer churn. The insights will help the business design targeted retention strategies and reduce customer attrition.

Problem Statement

A digital commerce platform is experiencing a high level of customer churn, resulting in reduced repeat purchases, lower customer retention, and a negative impact on long-term revenue growth. Although the platform has a strong customer base and order volume, management lacks clear visibility into the key reasons why customers stop purchasing or disengage from the platform.

Several business challenges currently exist:

- There is no clear understanding of which customer segments contribute the highest churn, especially across device usage, gender, city tier, payment mode, tenure group, and order frequency.
- The business cannot identify whether new customers or long-term customers are more likely to churn.
- The relationship between customer satisfaction score and churn is not clearly measured.
- It is unclear whether coupon usage improves retention or only temporarily delays churn.
- The platform does not know how inactivity, such as days since last order, affects customer churn risk.
- Customer engagement metrics such as hours spent on the app are not being used to understand why some highly active users still leave.
- High-value product categories such as Mobile Phones, Accessories, and Fashion show significant customer loss, but the reasons are not clearly identified.
- Customers using certain payment methods, especially Cash on Delivery, appear to churn more frequently, but this pattern has not been analyzed in detail.
- Distance from warehouse locations may influence delivery speed and customer satisfaction, but its impact on churn is not currently tracked.
- Without proper churn insights, the business cannot create targeted retention campaigns, personalized offers, loyalty programs, or customer recovery strategies.

To solve these challenges, this project uses Power BI to analyze customer churn across multiple dimensions, including customer demographics, satisfaction score, payment preferences, coupon usage, order behavior, inactivity period, engagement level, and tenure group. The goal is to identify the key drivers of churn and provide actionable recommendations to improve customer retention, increase repeat purchases, and strengthen long-term business growth.

Attribute (Column /Features) Details:

The dataset consists of customer-level information capturing demographic, behavioural, and transactional attributes. Below is a description of each column.

Column Name	Description
CustomerID	Unique identifier for each customer
Churn	Binary flag: 1 if customer has churned, 0 if still active
Tenure	Number of months the customer has been with the platform
PreferredLoginDevice	Device used most often to access the platform (e.g., Mobile, Tablet, Desktop)
CityTier	Tier classification of the customer's city (1, 2, or 3, where Tier 1 = most metropolitan)
WarehouseToHome	Distance (in km) from the nearest warehouse/delivery hub to the customer's home
PreferredPaymentMode	Customer's most frequently used payment method (e.g., Credit Card, Debit Card, UPI, COD)
Gender	Customer's gender (Male / Female / Other)
HourSpendOnApp	Average number of hours per week spent on the mobile app / platform
NumberOfDeviceRegistered	Total number of devices (mobile, tablet, desktop) registered to the customer's account
PreferedOrderCat	Customer's most preferred order category (e.g., Technology, Fashion, Grocery, etc.)
SatisfactionScore	Customer-reported satisfaction score (typically 1 = very dissatisfied to 5 = very satisfied)
MaritalStatus	Marital status (e.g., Single, Married, Divorced)
NumberOfAddress	Number of different delivery addresses associated with the customer's account
Complain	Binary flag: 1 if the customer has filed a complaint in the last year, else 0
OrderAmountHikeFromlastYear	Percentage increase or decrease in order amount compared to the previous year (can be negative)
CouponUsed	Binary flag: 1 if the customer has ever used a coupon, else 0
OrderCount	Total number of orders placed by the customer during their lifetime
DaySinceLastOrder	Number of days since the customer's most recent order
CashbackAmount	Total cashback amount (in local currency) earned by the customer

Tools & Technologies

- **MySql:** Data cleaning,Transformation
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Cleaning & Transformation:

1. Handle Missing Values

- **Numeric columns** (Tenure, WarehouseToHome, OrderAmountHikeFromlastYear, DaySinceLastOrder, HourSpendOnApp, CouponUsed, OrderCount):
  Replace NULL with the **mean** value of that column .
- **Categorical columns** (Gender, PreferredLoginDevice, PreferredPaymentMode, PreferedOrderCat, MaritalStatus):
  Replace NULL with the **mode** (most frequent value).
- **SatisfactionScore** (1–5 scale):
  Fill NULL with median (3) or mode.

2. Standardise Text Categories

- **PreferredLoginDevice**:
  Convert 'Phone' → 'Mobile Phone' (as already done).
- **PreferredPaymentMode**:
  Unify synonyms: 'CC' → 'Credit Card', 'COD' → 'Cash on Delivery', keep 'Debit Card', 'E wallet', 'UPI' as is.
- **PreferedOrderCat**:
  Ensure consistent capitalisation (e.g., 'Laptop & Accessory', 'Mobile', 'Fashion', etc.). Possibly map 'Mobile' to 'Mobile Phone' if needed.
- **MaritalStatus**:
  Standardise to 'Single', 'Married', 'Divorced' (no typos).

3. Fix Logical Inconsistencies

- **Tenure = 0 but OrderAmountHikeFromlastYear > 0**
  → Set Tenure to 1 (new customers shouldn't have a hike from last year).
  Tenure = 0 but OrderCount > 0
  → Could remain 0 (first order in same month). No change.
- **HourSpendOnApp = 0**
  → Replace with median value (e.g., 2) – more realistic than forcing 3.
- **WarehouseToHome = 0 or negative**
  → Treat as missing and replace with median.

4. Treat Outliers

- **WarehouseToHome**:
  Values above, say, 100 are unrealistic (distance in km/miles). Cap at 100 (or use IQR method). Original had 126, 127, etc. – likely data entry errors.
- **Tenure**:
  Cap at 60 months (5 years) – anything higher is improbable.
- **CashbackAmount**:
  If negative, set to 0.
- **OrderAmountHikeFromlastYear**:
  Extreme values (e.g., >100) could be capped at 100 or treated as outliers.

5. Ensure Correct Data Types & Ranges

- **Churn** and **Complain** should be BIT (0 or 1) – no action needed.
- **SatisfactionScore** must be between 1 and 5. Any value outside → set to nearest bound or median.
- **CityTier** should be 1, 2, or 3. Any other → set to mode (1).
- **NumberOfDeviceRegistered** – reasonable range (1–5).

6. Final Consistency Check

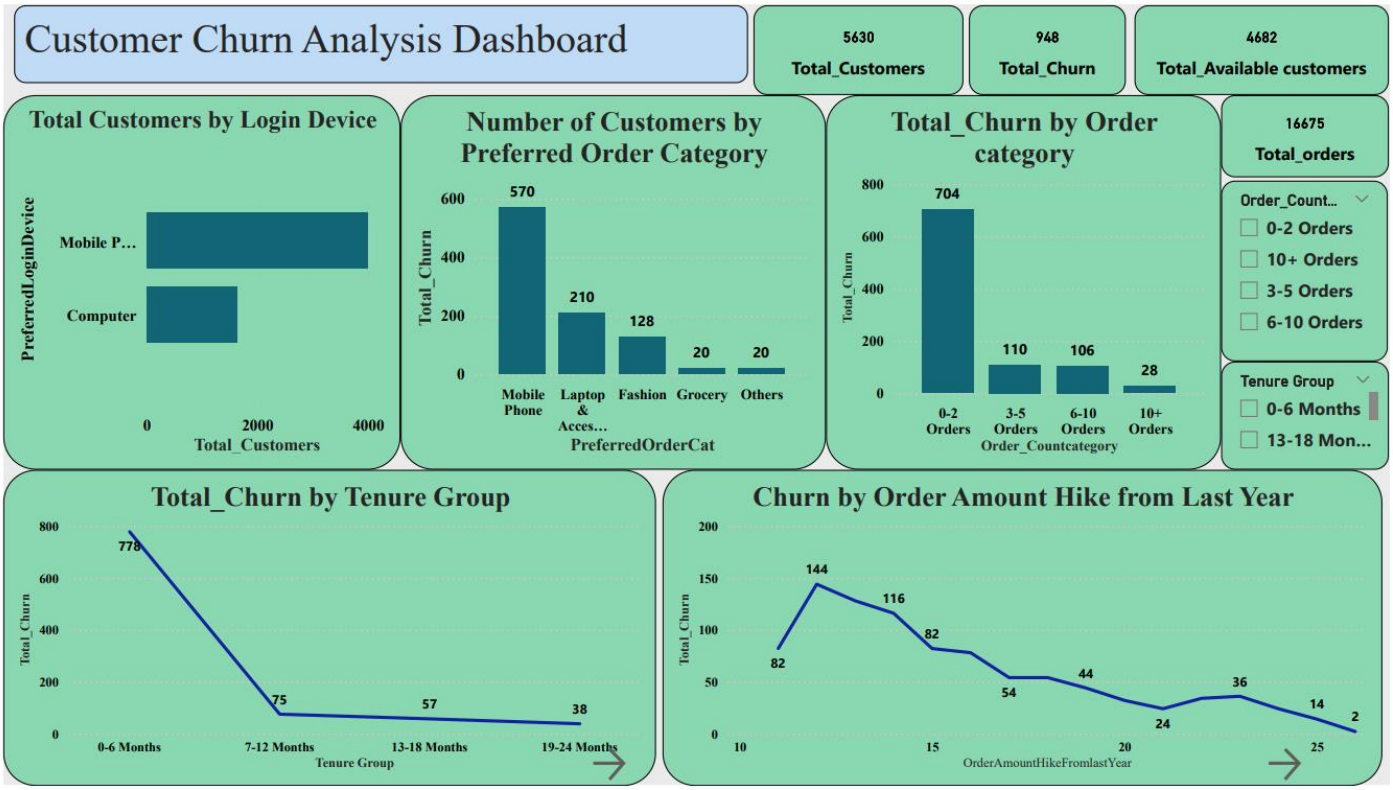
- Verify no NULL values remain in key columns.
- Check that Tenure is not negative.
- Confirm that DaySinceLastOrder is non-negative (0 = same day).

Data Transformation in Power BI

The following transformations were applied in Power Query / Power BI:

1. **Churn Rate Calculation**
   Churn Rate = SUM(Total_Churn) / SUM(Total_Customers)
   Shown as a percentage in visuals.
2. **Tenure Grouping**
   Tenure (in months) was binned into groups (e.g., 0–6, 6–12, 12–18, 18–24, 24+) to analyse churn by customer lifecycle stage.
3. **Days Since Last Order Binning**
   Created bins: 0–30 days, 31–60 days, 61–90 days, 90+ days to show how recency affects churn.

Visualizations Insights:



- Churn rate is high (~28%), indicating a significant loss of customers and revenue.
- Customers with 0–2 orders show the highest churn, meaning users drop off early.
- New customers (0–6 months tenure) contribute the most to churn.
- Mobile users dominate but also churn more, suggesting usability or performance issues.
- Customers with low order frequency churn more, while repeat users stay.
- Satisfaction scores 2 and 3 have highest churn, but even high scores show churn.
- High churn in Debit/Credit Card users indicates possible transaction issues.
- Customers with low coupon usage churn more, but discounts alone don't retain users.
- Mobile Phone and Fashion categories show higher churn.
- Users with high app usage still churn, indicating engagement without satisfaction.
- Inactive users (10+ days) are more likely to churn.
- Tier 1 cities show highest churn, likely due to competition.
- Overall churn is driven by low engagement, poor early experience, and satisfaction gaps .

Recommendations

- Implement a customer retention strategy as a top business priority.
- Focus on strong onboarding, first-purchase offers, and early engagement tactics.
- Introduce new user nurturing programs and personalized journeys.

- Improve mobile app UX, speed, and checkout flow.
- Encourage repeat purchases through loyalty programs and reminders.
- Strengthen feedback systems and proactive issue resolution.
- Optimize payment gateways and ensure smooth transactions.
- Use personalized and targeted offers instead of generic discounts.
- Improve product quality, pricing strategy, and category-specific offers.
- Focus on experience quality, not just time spent.
- Implement re-engagement campaigns (notifications, emails, offers)
- Enhance service quality, faster delivery, and competitive pricing in metro areas
- Implement automatic retention call-back within 24 hours for any customer who files a complaint.
- Create a 'Family Account' feature for users with 3+ addresses - offer shared cart, gift options, and household rewards.
- Offer 5% discount for registering a second device (tablet/desktop) to increase platform stickiness.
- Flag customers with negative order amount hike. Send win-back offer within 7 days of detection.
- For customers >30km from warehouse, offer 'Slower Delivery = Extra Cashback' option to manage expectations.
- Build a customer health score combining: complaint status + order amount trend + days since last order + device count. Alert retention team when score drops below threshold.

Conclusion

- The analysis reveals that customer churn is significantly high (~28%), indicating a major impact on business growth and retention.
- Churn is primarily driven by low customer engagement, especially among new users and customers with fewer orders.
- The early stage of the customer journey (first 6 months) is the most critical, where the majority of customers tend to drop off.
- Factors like poor mobile experience, low satisfaction levels, and inactivity further contribute to customer churn.
- Discounts and coupons alone are not sufficient, as customers still churn despite usage, highlighting deeper experience issues.
- High churn in specific categories and Tier 1 cities suggests the influence of competition and customer expectations.
- Overall, the analysis shows that customer experience, engagement, and satisfaction are the key drivers of retention, rather than just pricing or offers .

- By focusing on early engagement, personalization, improved user experience, and targeted retention strategies, the business can effectively reduce churn and improve long-term customer value.